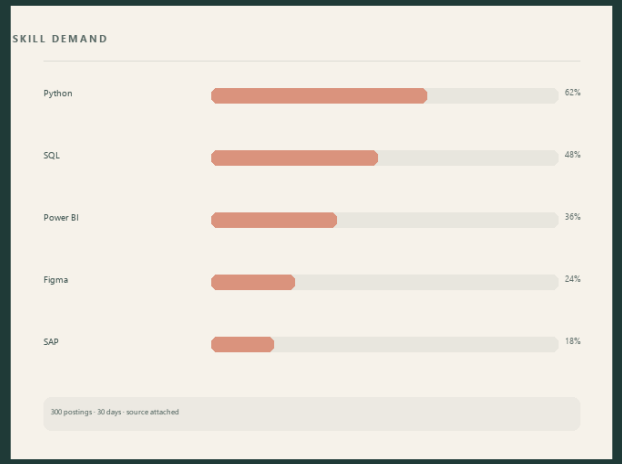
SDC HACKATHON 2026 · DOMAIN 1 · THE AIVERSECITY FUTURE-READY WORKFORCE DOMAIN

AURA-Workforce

Four checkpoints from the market evidence to a study plan a person can actually follow.



overinfinity · Ahmad Adam Hakimi bin Naeman

International Islamic University Malaysia

aura-workforce.replit.app

github.com/adamnaemman/sdc-hackathon

THE PROBLEM

Workers are told to reskill, but not what to do on Monday.

56%

of Malaysian professionals already use AI at work — only 26% have had formal training.

Hays Asia Salary Guide 2025

600,000+

workers projected to need reskilling within three to five years.

MOSTI

39%

of workers' core skills are expected to change by 2030.

WEF · Future of Jobs 2025

Yet the advice a worker receives is generic, static and rarely cites a source — while the training they are told to buy is expensive and often already out of date.

WHO IT IS FOR

Four people, one moment of need.

The displaced worker

A procurement assistant whose routine work is being automated. Needs a route out, not sympathy.

The switcher

A nurse, designer or clerk deciding where existing skills can carry them next.

HR and L&D

Planning next year's reskilling budget and needing evidence, not vendor decks.

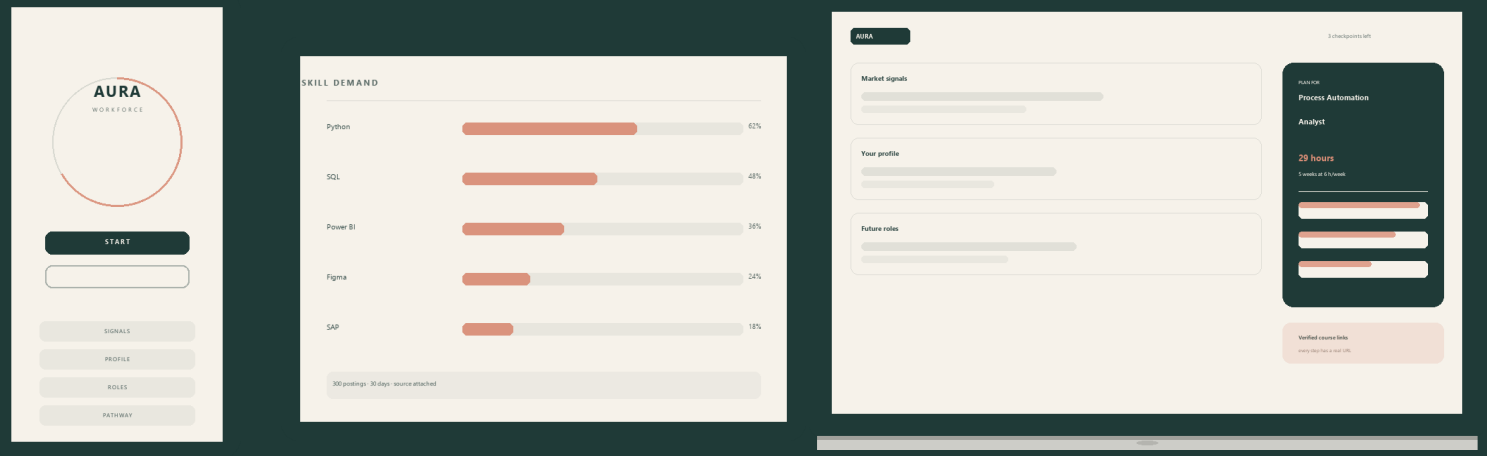
TVET and university advisers

Advising students from a printed list that was accurate five years ago.

The moment: the week a person learns their role is changing. They do not want a prediction — they want to know what to do next, and whether to trust it.

Personas derived from the Domain 1 brief

One page. Four checkpoints.



01 **Market signals**

Live demand from real postings, with the source and date.

02 **Your profile**

The AI expands what you write into skills you choose.

03 **Future roles**

Occupations proposed from your profile — not a fixed catalogue.

04 **Learning path**

A plan fitted to your weekly hours, with verified course links.

Same page on phone, tablet and desktop

Evidence, with its source attached.

Three hundred recent public postings, refreshed on demand.

The most requested skills plotted week by week — not a shortlist of our own roles.

Source, fetch time and sample size printed on the chart.

Six sourced labour-market signals from BLS and WEF, each with a source note.

The screen says what it is not: a global feed, not Malaysian vacancy counts.



Numbers that cannot be checked are not shown at all.

Name what you already bring.

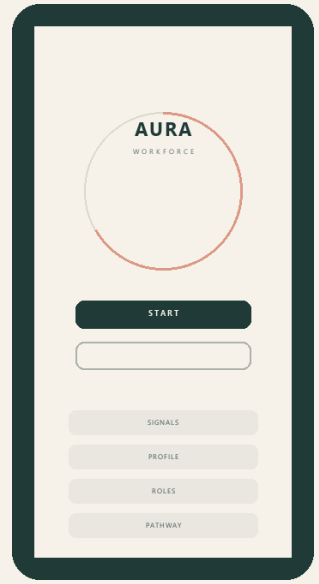
Role, industry and years of experience, then one model call.

The AI expands that into concrete, selectable skills and tools.

Tools are pre-ticked because they are factual; judgement skills stay unticked.

Custom skills survive, so a title we never modelled still counts.

Weekly hours entered here become the schedule in checkpoint 04.



Suggestions are offered, never applied on the person's behalf.

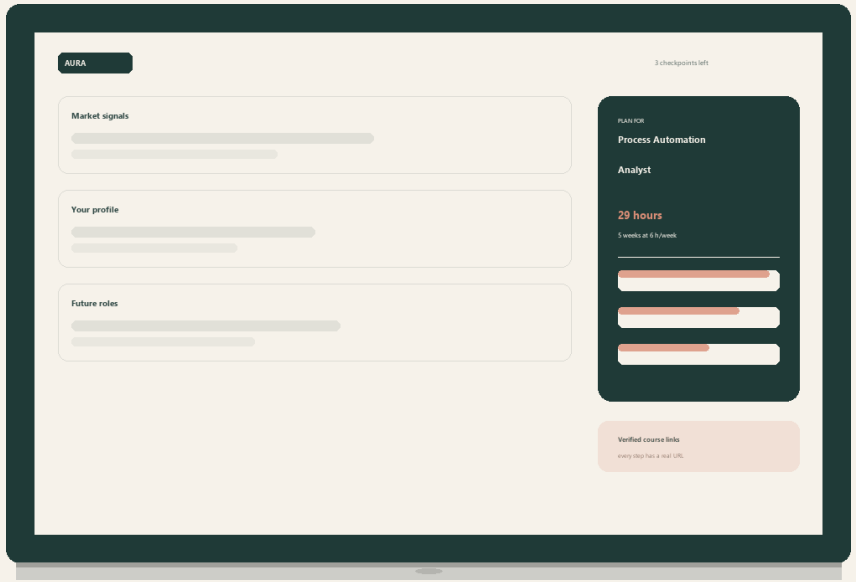
The roles the AI chose for you.

The model proposes occupations from the profile — including ones outside our catalogue.

Each comes with its reasoning, the skills to build, a first step, and a reference you can check.

No salary, growth figure or course link is shown for those occupations, because we could not source one.

They carry an explicit badge: exploratory, no verified pathway.



If the model cannot name a reference, the suggestion is discarded.

A plan built around the hours you actually have.

The model may only choose courses from a library we verified, link by link.

Every step shows provider, hours and the reason it was ordered there.

The plan is divided by the weekly hours the person said they have.

If nothing fits, it returns study topics with real search links — never an invented course.

Export to PDF, with a QR code on every step for the classroom or a printed handout.

Verified means fetched and checked before shipping.

TRUST, BY CONSTRUCTION

What the AI may not do.

Closed vocabulary

Every role and module identifier is validated against our catalogue; anything unknown is discarded before it reaches the screen.

Visible disagreement

The model's fit score is shown next to the deterministic estimate, so the two can be compared rather than blended.

No storage

No account, no database, no CV. The profile lives in the browser tab and is gone when it closes.

Injection treated as data

Free text is delimited, output is schema-validated, and the model never receives a tool that can write.

Each rule is a validation path in the repository — not a promise in a document.

The model chooses and explains; it never writes the catalogue.

Baseline, KPI, and a target we can be held to.

TARGETS

4.4.1 Relevant skills for employment

4.3.1 Access to lifelong learning

8.5.2 Employment and decent work

8.2.1 Productivity through technology

MEASUREMENT

Baseline — 56% use AI, 26% trained, 600k need reskilling.

KPI — share of recommendations carrying a verifiable source and date; time to a ranked plan.

Phase 2 target — 100% sourced, under five minutes, with a Malaysian trend source.

We report the KPI we can be measured on, not the one that flatters us.

Hardest risk: the model inventing a job or a course.

RISK AND FALLBACK

A generated role, course or statistic that does not exist.

- Catalogue and library are closed; unknown IDs are dropped server-side.
- The exploratory tier must name an occupational reference or the suggestion is removed.
- Course steps resolve only to URLs we fetched and verified before shipping.

Second risk: a global job feed is not Malaysian vacancy data — stated on screen, fixed in Phase 2.

STACK

React 19 · Vite · Tailwind 4

TanStack Query · Wouter

Express 5 · Zod · OpenAPI + Orval

DeepSeek (declared)

O*NET 30.1 · Arbeitnow

Replit hosting and deploy

Live prototype: aura-workforce.replit.app Repository: github.com/adamnaemman/sdc-hackathon

WHAT COMES NEXT

Phase 2: local data, sharper matching, proof.

Malaysian demand data	DOSM and local postings, so ranking reflects the country rather than a global feed.
Semantic skill matching	Embeddings instead of lexical overlap — the weakest part of the current engine.
Counsellor view	Cohort gaps and shared plans for TVET and university advisers.
Self-hosted model	RunPod credit already allocated; the client is OpenAI-compatible.